Executive Summary: Telco Customer Churn Analysis

1. Project Overview

Customer churn is a significant business challenge for telecommunications companies because losing existing customers can affect recurring revenue, customer lifetime value, and long-term business growth. Understanding why customers discontinue services can help companies identify customers who may be at risk and develop more effective retention strategies.

This project presents an Exploratory Data Analysis (EDA) of the Telco Customer Churn dataset using Python and its data analysis and visualization libraries, including Pandas, NumPy, Matplotlib, and Seaborn. The primary objective is to examine customer demographics, service subscriptions, contract types, tenure, and payment methods to identify patterns associated with customer churn.

The analysis uses descriptive statistics and visualizations, including count plots, a pie chart, a histogram, and stacked percentage bar charts, to explore customer behavior and compare churn patterns across different customer segments.

2. Dataset Overview and Data Preparation

The dataset contains **7,043 customer records and 21 columns**, covering customer demographics, account information, subscribed services, billing methods, charges, and churn status.

The dataset includes variables such as gender, senior-citizen status, tenure, contract type, Internet Service, Online Security, Tech Support, Payment Method, Monthly Charges, Total Charges, and Churn.

The following preprocessing and validation steps were performed:

- **Data structure:** Examined the dataset using `info()` and descriptive statistics to understand the data types and distributions.
- **Data type conversion:** Converted the `TotalCharges` column from a text-based format to a numeric format after replacing blank strings with zero.
- **Missing-value check:** The notebook reported zero null values after preprocessing.
- **Duplicate validation:** No duplicate customer IDs were identified.
- **Feature readability:** Converted the binary `SeniorCitizen` variable into Yes/No categories to improve interpretability in the visualizations.

These steps prepared the dataset for exploratory analysis. However, the blank values replaced with zero in `TotalCharges` should be validated against the original records to ensure that missing values are not being interpreted as actual zero charges.

3. Key Findings

3.1 Overall Customer Churn

The overall churn analysis indicates that **26.54% of customers discontinued their services**, while the remaining **73.46% stayed with the company**.

For the 7,043 customer records, this corresponds to approximately 1,869 churned customers and 5,174 retained customers.

The findings indicate that more than one in four customers in the dataset had discontinued their services. Although the majority of customers remained, the proportion of churned customers represents a meaningful retention challenge that warrants further investigation.

The overall churn distribution establishes the baseline against which customer segments can be compared. The subsequent analysis examines whether particular demographic characteristics, service subscriptions, contractual arrangements, or payment methods are associated with higher churn.

3.2 Customer Demographics and Gender

The analysis examined churn by gender and senior-citizen status to investigate whether demographic characteristics were associated with differences in customer retention.

Gender: The gender-based count plot compares churned and retained customers across male and female groups. The notebook does not establish a prominent gender-related pattern, so gender should not be treated as a major churn driver based on the current analysis alone.

Senior-citizen status: The analysis suggests that senior citizens have a comparatively higher proportion of churn than non-senior customers. This observation indicates that senior-citizen status may be a useful variable for further investigation.

However, the senior-citizen count plot and the percentage-based stacked bar chart serve different purposes. The count plot shows the number of customers in each group, while the normalized stacked chart compares the churn and retention percentages within each group.

The percentage-based comparison is particularly important because the dataset contains different numbers of senior and non-senior customers. Exact group-specific churn rates should be calculated before determining the size of the difference.

Business implication: If the difference is confirmed quantitatively, the company could investigate whether senior customers experience specific service, support, billing, or product-related difficulties that affect retention.

3.3 Customer Tenure and Retention

Customer tenure represents the length of time a customer has maintained a relationship with the company. The tenure histogram, separated by churn status, reveals a clear exploratory pattern: customers with shorter tenures are more likely to churn, while customers who have remained with the company for longer periods are more likely to stay.

The notebook particularly highlights customers who have used the services for only one or two months as a group with noticeable churn.

This finding suggests that the early stages of the customer lifecycle may represent an important retention opportunity. Customers who have recently subscribed may not yet have developed a strong relationship with the service provider or may be more willing to switch providers if their expectations are not met.

The analysis does not calculate exact churn percentages for individual tenure groups, so the magnitude of the relationship cannot yet be quantified.

Business implication: The company could investigate early customer experiences, including service activation, onboarding, technical support, billing clarity, and initial service satisfaction. Monitoring churn during the first few months could help identify retention opportunities earlier.

3.4 Contract Type and Customer Churn

Contract type is one of the most notable patterns identified in the notebook. The count plot comparing churn status across month-to-month, one-year, and two-year contracts indicates that customers on month-to-month contracts account for a substantially greater number of churned customers than customers on longer-term contracts.

Customers with one-year and two-year contracts appear comparatively more likely to remain with the company.

This pattern may reflect differences in contractual commitment, customer preferences, service satisfaction, or switching flexibility. However, the count plot alone cannot establish which of these explanations is responsible for the observed difference.

It is also important to distinguish the number of churned customers from the churn rate within each contract group. Because the number of customers in each contract category differs, comparing the percentage who churn within each group provides a more meaningful assessment of retention risk.

Business implication: The company could explore voluntary contract upgrades, loyalty benefits, and incentives for customers who prefer flexible contracts. Such offers should be based on customer needs and value rather than assuming that every month-to-month customer is dissatisfied.

3.5 Internet Services and Additional Service Subscriptions

The notebook examines nine service-related variables: Phone Service, Multiple Lines, Internet Service, Online Security, Online Backup, Device Protection, Tech Support, Streaming TV, and Streaming Movies.

The nine-panel count-plot visualization provides a consolidated view of churned and retained customers across these service categories.

Several patterns are highlighted in the analysis:

- **Online Security:** Customers without Online Security appear to have higher churn than customers who subscribe to the service.
- **Tech Support:** Customers without Tech Support also show a noticeably higher churn pattern, while customers with the service appear comparatively more likely to remain.
- **Internet Service:** Fiber-optic customers show relatively high churn in the exploratory comparison.
- **Other additional services:** Online Backup, Device Protection, Streaming TV, Streaming Movies, Phone Service, and Multiple Lines provide further opportunities to investigate differences in customer behavior.

These patterns suggest that service subscriptions may help explain differences in customer retention. However, the current notebook does not establish that subscribing to Online Security or Tech Support directly reduces churn. Customers who choose these services may differ from other customers in tenure, contract type, spending, or other characteristics.

Similarly, the relatively high churn observed among Fiber-optic customers warrants investigation but does not, by itself, establish dissatisfaction with Fiber-optic service.

Business implication: The company could examine service quality, customer support experiences, pricing, and the perceived value of additional services. It could also investigate whether particular combinations of services are associated with higher or lower churn.

3.6 Payment Method and Customer Churn

The payment-method analysis compares customer churn across electronic check, mailed check, bank transfer, and credit card payment categories.

The notebook identifies electronic check users as a group with a noticeable churn pattern. This makes payment method a useful variable for further investigation into customer retention.

However, the count plot displays customer counts rather than churn percentages within each payment category. The number of churned electronic check users alone does not establish that electronic check customers have the highest individual churn probability.

A more informative comparison would calculate the proportion of customers who churn within each payment method and compare those rates with the overall churn rate of 26.54%.

Business implication: If electronic check users demonstrate a higher churn rate after this comparison, the company could investigate payment-related issues, billing experiences, payment convenience, and the relationship between payment methods and other customer characteristics.

4. Consolidated Business Insights

The exploratory findings identify five areas that deserve particular attention:

1. **Early customer retention:** Short-tenure customers appear more vulnerable to churn, making onboarding and early engagement important areas for investigation.
2. **Contract flexibility:** Month-to-month customers show a more pronounced churn pattern than customers on longer-term contracts.
3. **Additional service subscriptions:** The absence of Online Security and Tech Support is associated with a noticeable churn pattern in the visualizations.
4. **Internet Service type:** Fiber-optic customers appear to have comparatively high churn, warranting investigation into service experience, pricing, and customer expectations.
5. **Payment behavior:** Electronic check users represent another customer segment worth examining through churn-rate comparisons.

The overall churn rate of 26.54% provides the baseline for these investigations. The notebook's visualizations help identify potential customer segments of interest, but exact differences between these segments should be quantified before prioritizing interventions.

5. Recommended Business Actions

Based on the exploratory patterns, the following actions could be considered:

Improve early-stage customer engagement. Develop an onboarding process that helps new customers understand their services, resolve activation issues, and access support during their initial months.

Investigate month-to-month customer retention. Examine the reasons these customers leave and evaluate whether flexible retention offers, loyalty benefits, or suitable longer-term plans could improve retention.

Evaluate the perceived value of additional services. Investigate whether customers without Online Security and Tech Support have different support experiences, satisfaction levels, or service requirements. Offer relevant services based on customer needs rather than assuming that additional subscriptions will automatically prevent churn.

Examine Fiber-optic customer experiences. Compare churn rates across Internet Service categories and investigate potential differences in pricing, reliability, support requests, and customer expectations.

Review payment-method patterns. Calculate churn rates by payment method and examine whether the electronic check pattern remains after accounting for contract type, tenure, and customer charges.

Introduce segment-level monitoring. Track churn rates across tenure bands, contract types, and service categories over time. This would help the business assess whether retention initiatives are associated with measurable improvements.

These recommendations should be treated as proposed actions for investigation and evaluation, not as interventions whose effectiveness has already been demonstrated by this analysis.

6. Limitations and Further Analysis

This project is an exploratory analysis rather than a causal study or a predictive modeling exercise. The visualizations identify associations and patterns, but they do not establish that any individual characteristic causes a customer to churn.

Most category-level visualizations in the notebook use customer counts. Because group sizes differ, count comparisons may be misleading when the objective is to assess churn risk. The next analytical step should calculate churn percentages within each category and compare them against the overall churn rate.

Further analysis should include:

- Calculating category-specific churn rates and percentage-point differences from the overall baseline.
- Examining tenure and monthly charges across churn groups.
- Investigating relationships between contract type, service subscriptions, payment methods, and churn.
- Applying statistical tests where appropriate to assess whether observed differences are likely to reflect more than random variation.
- Building classification models, such as Logistic Regression and Random Forest, to evaluate predictive performance using appropriate validation methods.
- Assessing precision, recall, F1-score, and other relevant metrics rather than relying on accuracy alone.

Any predictive model should be evaluated on held-out data, with preprocessing and feature selection performed in a way that avoids data leakage.

7. Conclusion

The Telco Customer Churn analysis provides an initial understanding of customer retention patterns across demographic, contractual, service, tenure, and payment-related variables. The overall churn rate of **26.54%** indicates that customer attrition is a meaningful business concern in the dataset.

The most notable exploratory patterns relate to short customer tenure, month-to-month contracts, the absence of Online Security and Tech Support, Fiber-optic Internet Service, and electronic check payments. These areas provide a practical starting point for more detailed investigation into customer behavior and potential retention opportunities.

By adding category-level churn percentages, validating the observed differences, and subsequently developing a machine learning model, this project can progress from descriptive visualization toward quantitative analysis and predictive customer retention. The ultimate objective is to help a telecommunications business identify higher-risk customer segments, understand the factors associated with churn, and make better-informed retention decisions.